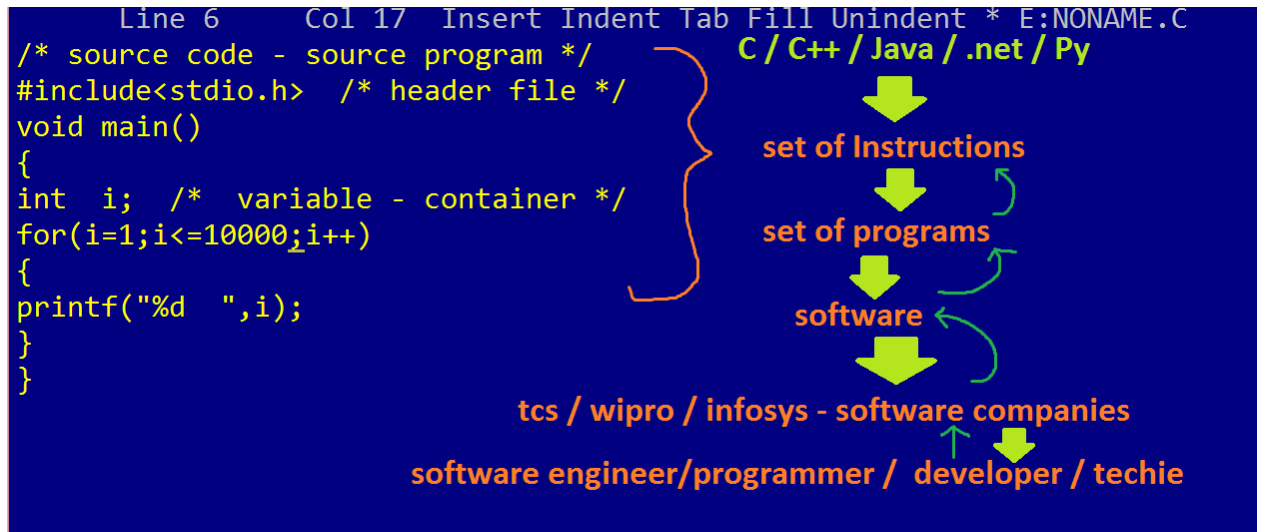


INTRODUCTION TO C

C is a

1. It is a high level / middle level programming language.

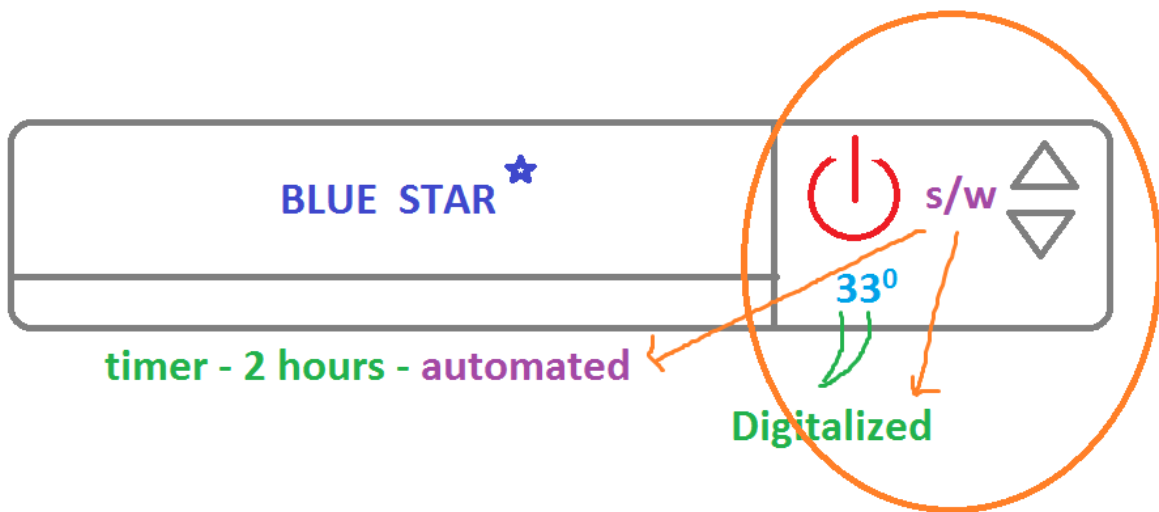
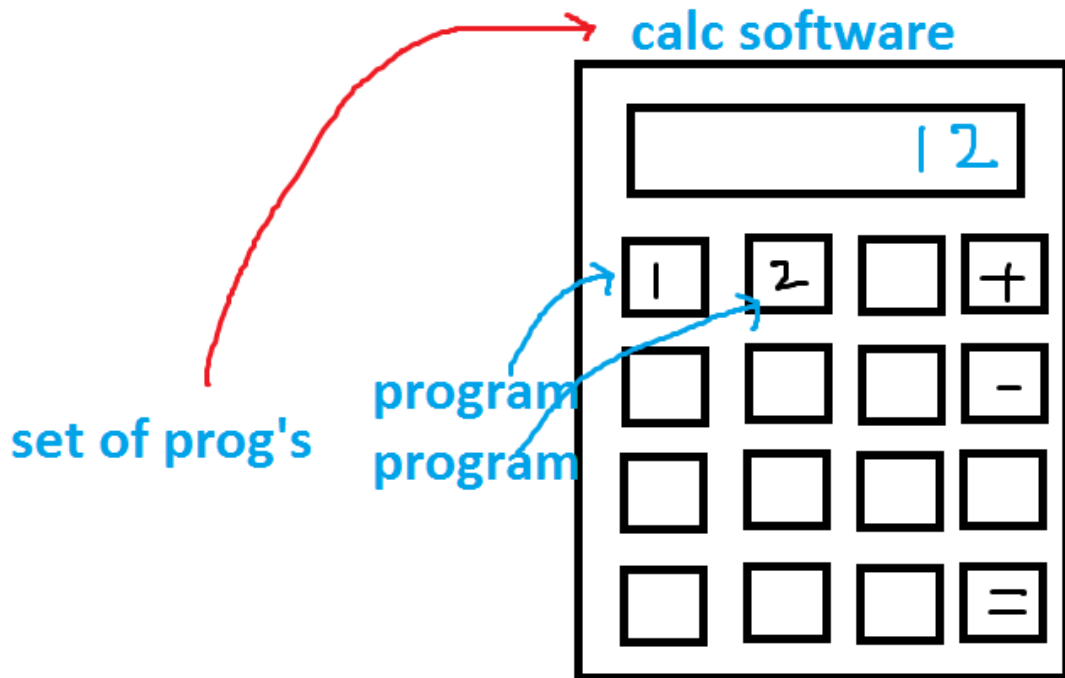


What is a program?

Set of instructions are called program

What is a software?

Set of programs is called software. As per IT Industry software is a digitalized and automated process.



We are having basically 2 type of software.

1.System software:

Eg: os, device drivers, translators

2.Application software

Eg: whatsapp, fb, insta,...

What is a language?

Generally the languages are used to communicate with others. For example the languages like telugu, English, hindi, Marathi etc are called human languages, which are used to communicate only with humans. But by using these languages we can't communicate with the machines. For that we are using the computer programming languages like C / C++ / java / Python / .net / Go / R language etc to create the programs [software]. These software making our work easy, faster and accurate.

Basically these languages are divided into 3 types.

1. Machine language: Created with binary code and very difficult to read by the user.

Eg: 10001111

2. Low level / assembly language: Created with English like shortcuts called **MNEMONICS**.

Eg: add, sub,...

Example for assembly programming:

The screenshot displays the Turbo C++ IDE interface. The top window, titled 'ASS.CPP', contains the following C code:

```
// example for low level / assembly program
#include<stdio.h>
void main() {
    int a = 10, b = 20, c;

    asm {
        mov ax,a
        mov bx,b
        add ax,bx
        mov c,ax
    }

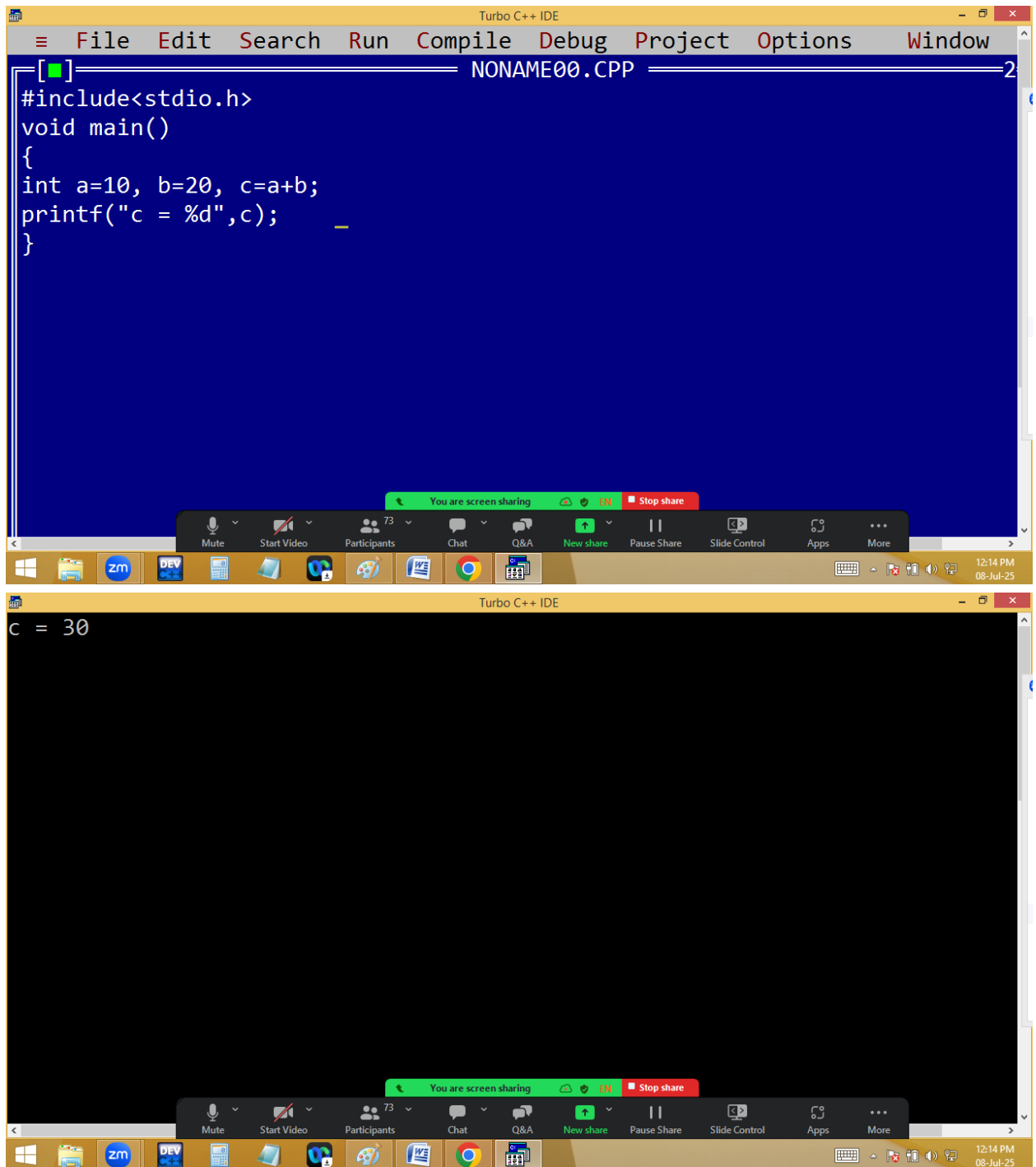
    printf("c = %d",c);
}
```

The bottom window shows the output of the program: 'c = 30'. The Windows taskbar at the bottom includes icons for various applications and the system clock, which shows 12:04 PM on 08-Jul-25. A green status bar at the bottom of the IDE windows indicates 'You are screen sharing'.

3. High level language: Created with simple English and easy to understand.

Eg: addition, subtraction,...

Example for high level program:



The image consists of two screenshots of the Turbo C++ IDE. The top screenshot shows the source code of a C program in a file named NONAME00.CPP. The code is as follows:

```
#include<stdio.h>
void main()
{
int a=10, b=20, c=a+b;
printf("c = %d",c);
}
```

The bottom screenshot shows the output of the program, which is "c = 30". Both screenshots include a taskbar at the bottom with various application icons and a system tray showing the time as 12:14 PM on 08-Jul-25. A screen sharing overlay is visible in the center of both screenshots, indicating that the content is being shared.

C is a **high level language** with **low level features**.
Hence it is a **middle level language**.

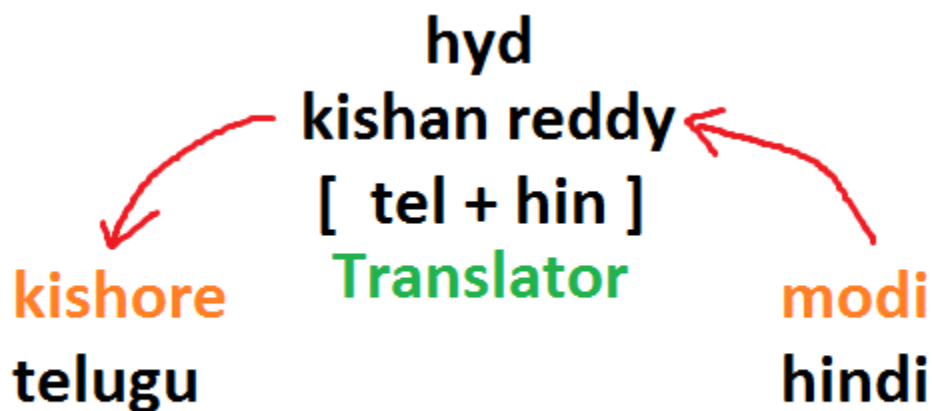
C low level features are used to design system software.

C high level features used to design application software.

Hence C is a **Multi-Purpose programming language**.

2. C is a compiler based programming language.

What is a translator?



Always the user given instructions are in English, which are called source code / source program. but the computer is not able to understand this English. To convert this English code to binary code

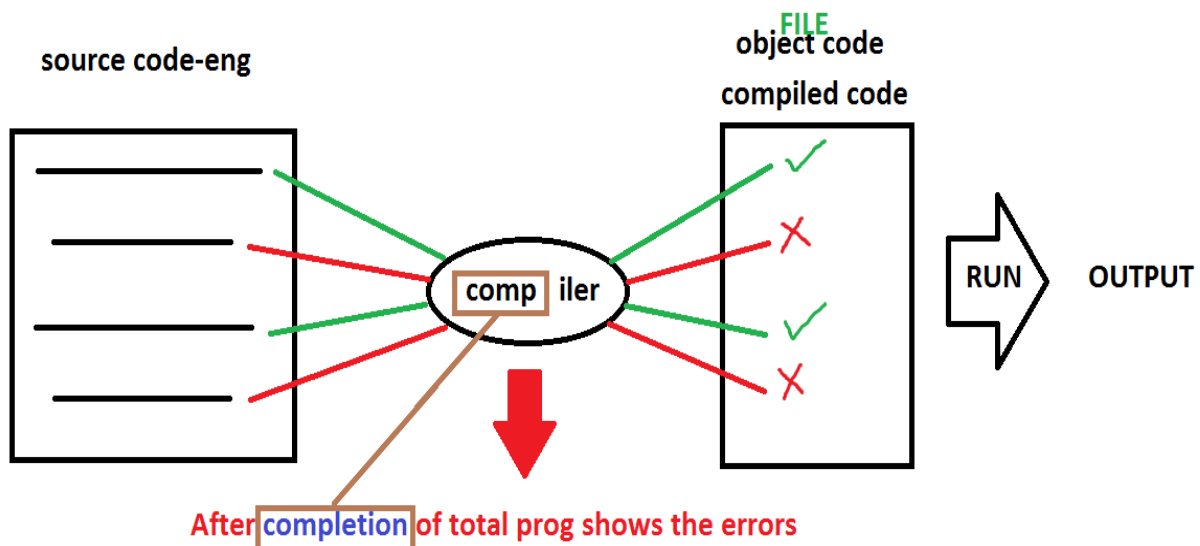
and to check the errors we are using the translators.

We are having 3 types of translators.

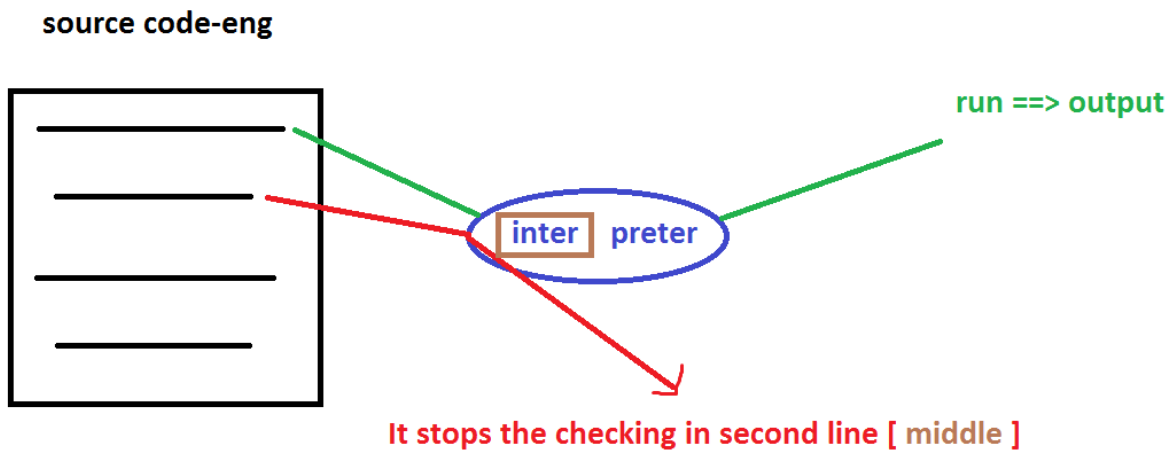
- 1.Compiler
- 2.Interpreter
- 3.Assembler

Compiler and interpreter both used to convert high level programs to binary code.

Compiler converts the total source code into binary code **at once** by leaving error lines.



Interpreter converts **line by line**.



Assembler is used to convert low level programs to binary code.

Assembler working style is similar to compiler.

In C & C++ we are using only the compilers. Hence they are called compiler based programming languages.

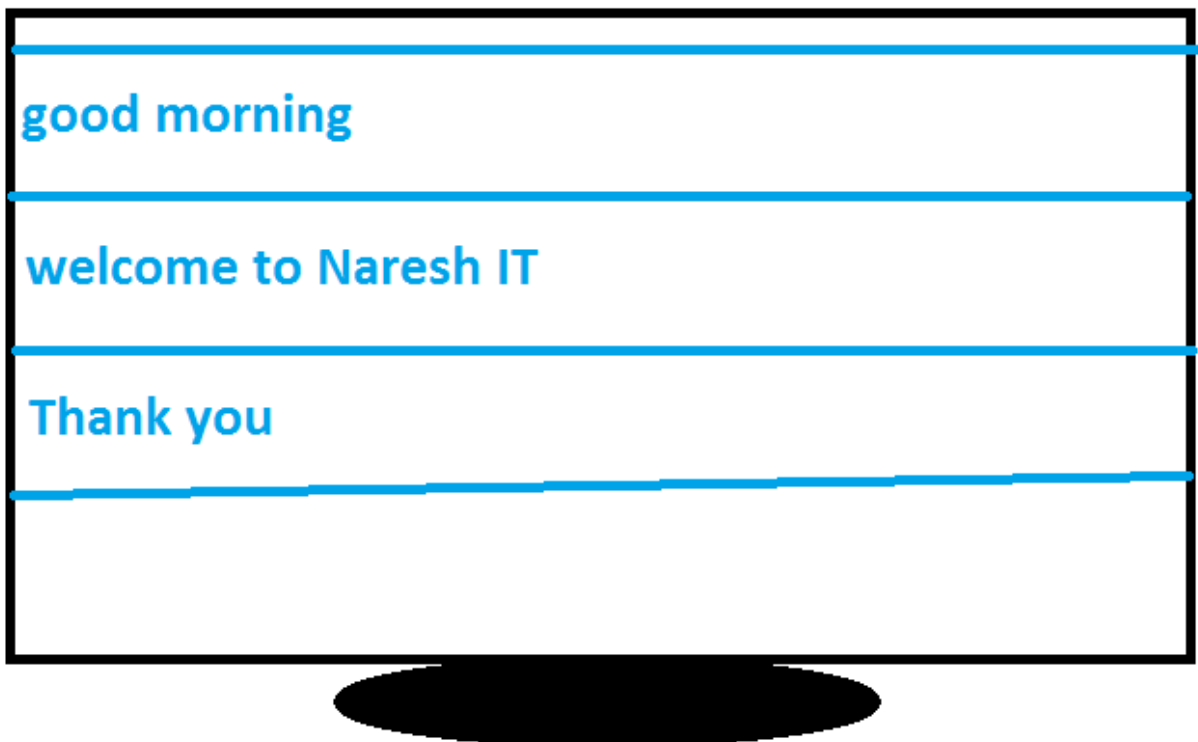
In java / .net / py etc we are using both compiler and interpreters. Hence they are called compiler based and interpreted languages.

3. C is a procedure oriented programming language [POPs]

What is called programming paradigm?

Every programming language comes with a particular syntax [rules and regulations] and a structure, which is technically called programming paradigm. To solve a problem, we should have to use that **programming paradigm** only.

Before C language the languages are using **monolithic programming paradigm**.



The image shows a screenshot of a Turbo C++ IDE window. The top menu bar includes File, Edit, Search, Run, Compile, Debug, Project, Options, and Window. The main editor area displays a C program named NONAME00.CPP. The code is as follows:

```
// monolithic programming
#include<stdio.h>
void main()
{
printf("-----\n");
printf("Good morning\n");
printf("-----\n");
printf("Welcome to Naresh IT\n");
printf("-----\n");
printf("Thank You\n");
printf("-----");
}
```

Below the editor, a status bar shows the line number 1:27. The output window at the bottom displays the program's execution results:

```
-----
Good morning
-----
Welcome to Naresh IT
-----
Thank You
-----
```

The Windows taskbar at the bottom shows the time as 12:22 PM on 09-Jul-25. A Zoom toolbar is also visible, indicating a screen sharing session.

POPs – Procedure oriented programming structure:

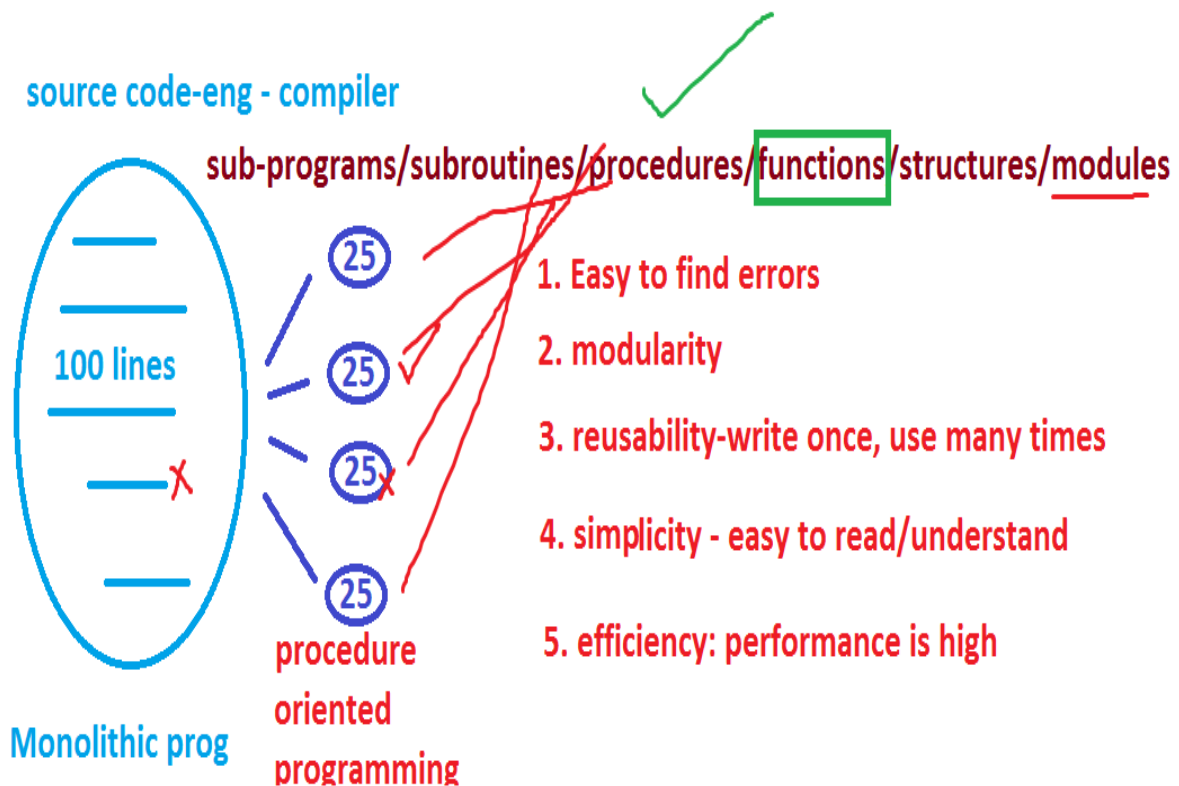
The image shows a screenshot of a Turbo C++ IDE window. The top menu bar includes File, Edit, Search, Run, Compile, Debug, Project, Options, and Window. The main editor area displays a C program named NONAME00.CPP. The code defines a procedure-oriented programming function 'line()' that prints a dashed line, and a 'main()' function that calls 'line()' to print 'Good morning', 'Welcome to Naresh IT', and 'Thank You'. Below the editor, a console window shows the program's output: three dashed lines, each followed by one of the printed messages. The Windows taskbar at the bottom shows various application icons, including Zoho, DEV, and Google Chrome. A Zoom toolbar is also visible, indicating a screen share session.

```
File Edit Search Run Compile Debug Project Options Window
[ ] NONAME00.CPP 1
// procedure oriented programming
#include<stdio.h>
void line() // procedure or function
{
printf("-----\n");
}
void main() // procedure or function
{
line();
printf("Good morning\n");
line();
printf("Welcome to Naresh IT\n");
line();
printf("Thank You\n");
line();
}
```

Good morning

Welcome to Naresh IT

Thank You



Basic language following monolithic programming.

C is following procedure oriented programming structure [POPs]

C++ and Python following POPs and OOPs. Hence they are called multi paradigm programming languages.

Java and .net are oops

